MS&E 228 (Winter 2026) — Lecture 15 Notes

Instrumental Variables

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Readings. *Applied Causal Inference Powered by ML and AI*, Chapter 13. Key papers: Angrist, Imbens, & Rubin (1996), Card (1995), Staiger & Stock (1997).
Jupyter Notebook. DML for IV on 401(k) Data.

1 Introduction: The Instrumental Variable Idea

In the previous chapter, we confronted the challenge of unobserved confounding and explored two distinct paths forward. The first, sensitivity analysis, provided a way to quantify the potential impact of an unobserved confounder, allowing us to bound the true treatment effect. The second, the front-door criterion, offered an alternative identification strategy that could succeed even when conditional exogeneity failed. This chapter introduces a third, and arguably the most influential, strategy for dealing with unobserved confounding: **instrumental variables (IV)**.

The IV approach is qualitatively similar to the front-door method in that it leverages a specific causal structure to achieve identification, but its mechanics are very different and it is far more widely used in applied research. The goals for this chapter are to:

1. Introduce the core idea of an **instrumental variable** and how it can be used to isolate a causal effect in the presence of unobserved confounding.
2. Define a new causal estimand, the **Local Average Treatment Effect (LATE)**, and the subpopulation of **compliers** for whom this effect is identified.
3. Show how to **estimate** IV models using analogs of the Doubly Robust and Double/Debiased Machine Learning methods we have developed.
4. Discuss the practical problem of **weak instruments** and introduce methods for robust inference.

1.1 Motivating Example: The Eager Trial Revisited

We return to the Eager trial, a study designed to understand the effect of low-dose aspirin (D) on pregnancy rates (Y). In this trial, patients were randomly assigned to a treatment group ($Z = 1$), which was recommended to take aspirin, or a control group ($Z = 0$), which was given a placebo. As we discussed in previous lectures, a key challenge was **non-compliance**: not all participants adhered to their assigned treatment. Some in the treatment group did not take the aspirin.

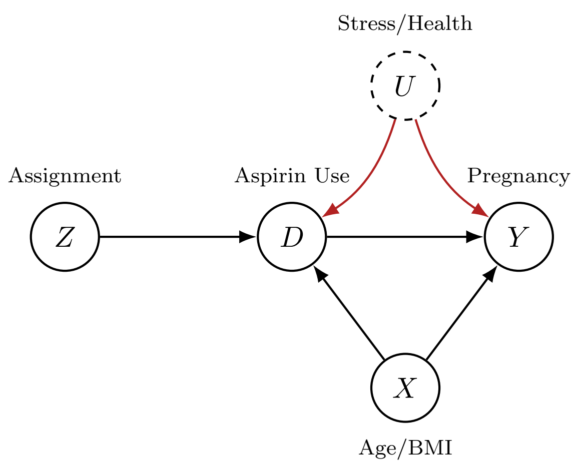
When we first analyzed this trial, we used the doubly robust estimator, controlling for a set of observed patient characteristics X (e.g., age, BMI) to account for confounding in the compliance decision. However, it is highly plausible that there are *unobserved* factors that influence both compliance and the outcome. For instance, a patient's stress level or general health habits (U), which were not measured, could affect their decision to take the pills and also have a direct impact on their pregnancy outcome. This introduces unobserved confounding.

How was Adherence Measured in the Eager Trial?

In the Eager trial, adherence was partially measured using pill bottle weights at follow-up visits. Based on these weight measurements, the physician's could judge whether the patient had taken a sufficient amount of pills, to constitute "taking the treatment".

1.2 The IV Causal Graph

This situation is captured by the canonical instrumental variable causal graph, shown below.



The random assignment Z is called the **instrument**. The key features of this graph are:

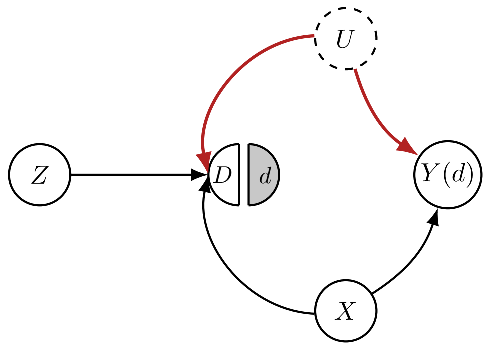
1. The instrument Z affects the treatment D. *(In the context of the trial this states that being recommended to take aspirin, had an effect of whether you would take aspirin or not)*

2. The instrument Z does not have a direct arrow to the outcome Y. Its only causal pathway to Y is mediated through the treatment D. This is known as the **exclusion restriction**. *(In the context of the Eager trial, this states that being recommended to take aspirin has no effect on your outcome, if that recommendation didn't change your decision to take aspirin or not)*

3. There is an unobserved confounder U that affects both D and Y, creating a backdoor path $D \leftarrow U \rightarrow Y$. *(In the context of the Eager trial, these are the unobserved reasons that*

led someone to not take their aspirin pills, and which could also correlate with pregnancy outcomes, e.g., stress levels)

Because of the open backdoor path through U, the conditional exogeneity assumption $Y(d) \perp\!\!\!\perp D \mid X$ fails. We can see this clearly by drawing the Single World Intervention Graph (SWIG) for the intervention $\text{fix}(D = d)$.



Backdoor Path is Open

The path $D \leftarrow U \rightarrow Y(d)$ is open because U is unobserved. The conditional exogeneity property $Y(d) \perp\!\!\!\perp D \mid X$ **fails**. We cannot use our standard DR/DML estimators to identify the effect of D on Y.

The central question of this chapter is: can we cleverly use the random assignment Z to bypass the unobserved confounder and still identify a causal effect of D on Y?

2 Identification with Instrumental Variables

To identify the effect of D on Y, we follow a logic similar to the front-door formula: express the target causal effect as a function of other effects that *are* identifiable from the data. In the IV graph, which causal effects can we identify using standard methods?

Because the instrument Z is randomized, it is exogenous. Therefore, we can identify the causal effect of Z on any other variable in the system. Specifically, two effects are of interest:

- The effect of the instrument on the treatment:

$$\alpha = \text{ATE}(Z \rightarrow D) = \mathbb{E}[D(Z = 1) - D(Z = 0)] = \mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0].$$

  This measures the degree of compliance with the assignment.

- The effect of the instrument on the outcome:

$$\beta = \text{ATE}(Z \rightarrow Y) = \mathbb{E}[Y(Z = 1) - Y(Z = 0)] = \mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0].$$

This is the famous **Intent-to-Treat (ITT)** effect, as it measures the effect of the "recommendation" or the "intention-to-treat" and not the actual treatment.

Our target quantity is the Average Treatment Effect (ATE) of aspirin on pregnancy, $\gamma = \mathbb{E}[Y(D = 1) - Y(D = 0)]$. Given that the only path from Z to Y is through D, it is tempting to assume a simple multiplicative relationship: $\beta = \alpha \cdot \gamma$. If this were true, we could solve for our target effect:

$$\gamma = \frac{\beta}{\alpha} = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0]}$$

This is the classic **IV estimator**, often called the **Wald estimator**. It has a nice interpretation: we take the overall effect of the recommendation on the outcome (the ITT effect) and rescale it by the compliance rate. The ITT effect is a diluted version of the true treatment effect, and we divide by the compliance rate α (a number between 0 and 1) to inflate it back to the full effect.

2.1 The Catch: Treatment Effect Heterogeneity

However, the simple logic $\beta = \alpha \cdot \gamma$ is flawed. It implicitly assumes that the average of a product is the product of the averages, which is generally not true. To see why, we must decompose the effect at the individual level. For any individual, the causal chain holds due to the exclusion restriction:

$$\begin{aligned} Y(Z=1) - Y(Z=0) &= \underbrace{Y(D(Z=1)) - Y(D(Z=0))}_{\text{exclusion restriction}} \\ &= \underbrace{(D(Z=1) - D(Z=0))}_{\text{Individual Compliance}} \times \underbrace{(Y(D=1) - Y(D=0))}_{\text{Individual Treatment Effect (ITE)}} \end{aligned}$$

When we average this over the population, we get:

$$\beta = \mathbb{E}[Y(Z=1) - Y(Z=0)] = \mathbb{E}\Big[(D(Z=1) - D(Z=0)) \cdot (Y(D=1) - Y(D=0))\Big]$$

This is not, in general, equal to $\mathbb{E}[D(Z = 1) - D(Z = 0)] \cdot \mathbb{E}[Y(D = 1) - Y(D = 0)] = \alpha \cdot \gamma$.

The Problem of Heterogeneity

The simple formula $\beta = \alpha \cdot \gamma$ only holds if we can pull the ITE term, $Y(D = 1) - Y(D = 0)$, outside the expectation. This is only possible if the individual treatment effect is independent of the individual compliance event, i.e., the drivers of treatment effect heterogeneity, are independent of the drivers of compliance. For instance, this would hold if the treatment effect was **homogeneous**—that is, if it is a constant γ for all individuals. If the effect of aspirin is different for different people (treatment effect heterogeneity), then there may be a correlation between a person's compliance behavior and their individual treatment effect. For example, people who are more likely to comply might also be those who benefit more (or less) from the treatment. This correlation prevents the simple decomposition.

2.2 The Role of Homogeneity

For many decades, the IV formula was justified by assuming constant treatment effects. Under this assumption, the treatment effect γ is the same for all individuals, so it can be pulled out of the expectation, and the simple ratio β/α recovers the ATE. One might try to relax this slightly by assuming **conditional homogeneity**:

$$Y(D=1) - Y(D=0) \perp\!\!\!\perp D(Z=1) - D(Z=0) \mid X$$

i.e., the treatment effect is independent of the compliance within strata defined by X. In this case, one could run the IV analysis separately within each stratum of X

$$\begin{aligned} \text{CATE}(X) = \mathbb{E}[Y(D=1) - Y(D=0) \mid X] &= \frac{\mathbb{E}[Y(Z=1) - Y(Z=0) \mid X]}{\mathbb{E}[D(Z=1) - D(Z=0) \mid X]} \\ &= \frac{\mathbb{E}[Y \mid Z=1, X] - \mathbb{E}[Y \mid Z=0, X]}{\mathbb{E}[D \mid Z=1, X] - \mathbb{E}[D \mid Z=0, X]} \end{aligned}$$

and then average the results:

$$\text{ATE} = \mathbb{E}[\text{CATE}(X)] = \mathbb{E}\left[\frac{\mathbb{E}[Y \mid Z=1, X] - \mathbb{E}[Y \mid Z=0, X]}{\mathbb{E}[D \mid Z=1, X] - \mathbb{E}[D \mid Z=0, X]}\right]$$

However, this still requires the strong assumption that treatment effect heterogeneity within groups defined by X, is uncorrelated with compliance, which is often implausible.

2.3 The LATE Theorem: Re-interpreting the IV Ratio

Instead of assuming homogeneity, the seminal work of Guido Imbens, Joshua Angrist and Donald Rubin, asked a different question: without any such homogeneity assumption, what does the IV ratio β/α actually estimate? Their Nobel Prize-winning insight was that, under a much weaker assumption, the ratio identifies the average treatment effect for a specific, policy-relevant subgroup of the population.

To understand this, we first need to introduce the **monotonicity** assumption.

Assumption: Monotonicity (No Defiers)

For every individual, the instrument can only encourage, not discourage, treatment uptake. Formally, for all individuals, $D(Z=1) \geqslant D(Z=0)$.

This assumption rules out the existence of "defiers," people who would do the opposite of what they are assigned (i.e., take aspirin only if assigned to placebo, and vice-versa). Under monotonicity, the population can be divided into three types based on their potential response to the instrument. The following table summarizes the classification:

Type	$D(Z=0)$	$D(Z=1)$	$D(1)-D(0)$
Complier	0	1	1
Always-Taker	1	1	0
Never-Taker	0	0	0
Defier (ruled out)	1	0	-1

- **Compliers:** People who take the treatment if and only if they are assigned to it. For them, $D(Z=1)=1$ and $D(Z=0)=0$, so $D(Z=1)-D(Z=0)=1$.
- **Always-Takers:** People who always take the treatment, regardless of assignment. For them, $D(Z=1)=1$ and $D(Z=0)=1$, so $D(Z=1)-D(Z=0)=0$.
- **Never-Takers:** People who never take the treatment, regardless of assignment. For them, $D(Z=1)=0$ and $D(Z=0)=0$, so $D(Z=1)-D(Z=0)=0$.

Notice that the individual compliance effect term, $D(Z=1)-D(Z=0)$, is equal to 1 for compliers and 0 for everyone else. It is an indicator for being a complier. Now we can return to our decomposition of the ITT effect:

$$
\begin{aligned}
\beta &= \mathbb{E}\Big[(D(Z=1)-D(Z=0))\cdot(Y(D=1)-Y(D=0))\Big] \\
&= \mathbb{E}\Big[\mathbf{1}\{\text{is Complier}\}\cdot(Y(D=1)-Y(D=0))\Big] \\
&= \mathbb{E}[Y(D=1)-Y(D=0)\mid \text{is Complier}]\cdot P(\text{is Complier})
\end{aligned}
$$

where the last step follows from the law of total expectation. What is the probability of being a complier? It is simply the expectation of the complier indicator:

$$
P(\text{is Complier}) = \mathbb{E}[\mathbf{1}\{\text{is Complier}\}] = \mathbb{E}[D(Z=1)-D(Z=0)] = \alpha
$$

Substituting this back into our equation for β, we get:

$$
\beta = \mathbb{E}[Y(1)-Y(0)\mid \text{is Complier}]\cdot\alpha
$$

Rearranging this gives the celebrated LATE theorem.

The Local Average Treatment Effect (LATE) Theorem

Under the assumptions of instrument validity (exogeneity and exclusion) and monotonicity, the IV ratio identifies the Average Treatment Effect for the subpopulation of compliers:

$$
\frac{\text{ATE}(Z\to Y)}{\text{ATE}(Z\to D)} \triangleq \frac{\beta}{\alpha} = \mathbb{E}[Y(D=1)-Y(D=0)\mid \text{is Complier}]
$$

This quantity is known as the Local Average Treatment Effect (LATE).

This is a landmark result. It tells us that even with unobserved confounding and heterogeneous treatment effects, the simple IV ratio still has a valid causal interpretation. It is not the ATE for the whole population, but the ATE for the subpopulation of individuals whose treatment choice is actually influenced by the instrument. In many policy contexts, this is precisely the group of interest. For instance, if a job training program is offered, the LATE is the effect on those who would choose to participate because the program was offered. This is the relevant effect for deciding whether the program is worthwhile, assuming the program cost is only incurred for participants.

Policy Interpretation of the LATE

One important subtlety is that the IV approach cannot tell us the effect for always-takers or never-takers, because the instrument does not change their treatment status. The LATE is "local" to the compliers. A common criticism is that the complier subpopulation is instrument-specific and may not generalize. However, the counterargument is compelling: it is better to have a correct estimate of the causal effect for *some* well-defined subpopulation than a biased estimate of the effect for the whole population. An observational study that ignores unobserved confounding gives the wrong answer for everyone; IV gives the right answer for the compliers.

2.4 Generalizing the IV Approach

The LATE formula is not limited to simple RCTs with non-compliance. The core logic applies whenever we can identify an instrument Z that affects the outcome Y only through the treatment D (exclusion restriction). In any such case, under the monotonicity restriction (instrument can only encourage treatment), we can always write:

$$\text{LATE} = \frac{\text{ATE}(Z \to Y)}{\text{ATE}(Z \to D)}$$

and under the homogeneity restriction (effect heterogeneity is independent of compliance), we can always write:

$$\text{ATE} = \frac{\text{ATE}(Z \to Y)}{\text{ATE}(Z \to D)}$$

Subsequently, all we need is a method to identify the effect of the instrument Z on the treatment D and the outcome Y. If for instance the instrument is conditionally exogenous, conditional on a set of observable factors X, then we can identify the effect of Z on D and Z on Y by conditioning on X and applying the g-formula. This allows us to use IV in a much broader range of settings.

- **Conditionally Randomized Trials:** In many experiments, the assignment probability itself depends on observed covariates X. For example, in the Eager trial, the randomization could have been stratified by age. In this case, the instrument Z is only exogenous conditional on X. We can still identify the conditional effects needed for the LATE ratio by applying the g-formula or DR-estimators for the numerator and denominator, averaging over X.

- **Observational Studies:** The IV framework extends to purely observational data, provided we can find an "observational instrument"—a variable that is plausibly conditionally exogenous and satisfies the exclusion restriction. Finding a valid observational instrument is a creative and challenging part of applied econometrics.

Famous Observational Instruments

Some of the most influential studies in economics are based on clever observational instruments:

- **Quarter of birth** was used by Angrist and Krueger (1991) as an instrument for the effect of years of schooling on earnings. Compulsory schooling laws require students to stay in school until a certain age, so those born earlier in the year can legally drop out with slightly less schooling than those born later.
- **Distance to college** was used by Card (1995) as an instrument for the effect of education on earnings. Living closer to a college reduces the cost of attendance and thus encourages enrollment, but is plausibly uncorrelated with innate ability after controlling for other factors.
- **Weather patterns** in a production region can serve as instruments for the effect of price on demand. For example, a drought in Colombia affects the cost of coffee beans, which in turn affects the price of coffee in the US, but the Colombian weather has no direct effect on US consumer demand.

In all these cases, the instrument is only valid after conditioning on a rich set of control variables X. This brings us to the problem of estimation in the presence of high-dimensional controls.

3 Estimation of LATE with High-Dimensional Controls

Having established that the LATE is an identifiable causal quantity, we now turn to its estimation. In the simplest case where the instrument Z is fully randomized and there are no other covariates X, the LATE is estimated by the simple ratio of differences-in-means. However, as we discussed, in most realistic settings (including observational studies), the instrument is only exogenous after conditioning on a potentially high-dimensional set of controls X. In this case, the numerator and denominator of the LATE can be identified via g-formulas:

$$\text{LATE} = \frac{\mathbb{E}_X[\mathbb{E}[Y \mid Z = 1, X] - \mathbb{E}[Y \mid Z = 0, X]]}{\mathbb{E}_X[\mathbb{E}[D \mid Z = 1, X] - \mathbb{E}[D \mid Z = 0, X]]}$$

This is a ratio of two g-formula-style quantities. As we have learned, a simple plug-in approach—where we train machine learning models for the nuisance functions (e.g., $\mu_1(x) = \mathbb{E}[Y \mid Z = 1, X = x]$) and then average their predictions—can suffer from regularization bias and complex, non-normal limiting distributions. To achieve $\sqrt{n}$-consistent and asymptotically normal estimation, we need a doubly robust approach.

3.1 A Doubly Robust Estimator for LATE (DR-LATE)

We can construct a doubly robust estimator for the LATE by constructing separate DR estimators for the numerator and the denominator. Let the nuisance functions be:

- $\mu_z(x) = \mathbb{E}[Y \mid Z = z, X = x]$ (Outcome regression for each instrument arm)
- $m_z(x) = \mathbb{E}[D \mid Z = z, X = x]$ (Treatment regression for each instrument arm)
- $p(x) = P(Z = 1 \mid X = x)$ (Propensity score of the instrument)

The DR score for the numerator (the ITT on Y) is:

$$\psi_Y(Y, Z, X) = (\mu_1(X) - \mu_0(X)) + \frac{Z}{p(X)}(Y - \mu_1(X)) - \frac{1-Z}{1-p(X)}(Y - \mu_0(X))$$

And the DR score for the denominator (the ITT on D) is:

$$\psi_D(D, Z, X) = (m_1(X) - m_0(X)) + \frac{Z}{p(X)}(D - m_1(X)) - \frac{1-Z}{1-p(X)}(D - m_0(X))$$

The DR-LATE estimator is the ratio of the sample means of these two scores.

The DR-LATE Estimator

Let $\hat{\psi}_Y$ and $\hat{\psi}_D$ be the DR scores computed using cross-fitted estimates of the nuisance functions. The DR-LATE estimator is:

$$\hat{\theta}_{\text{LATE}} = \frac{\mathbb{E}_n[\hat{\psi}_Y]}{\mathbb{E}_n[\hat{\psi}_D]}$$

This estimator is $\sqrt{n}$-consistent and asymptotically normal, provided the nuisance models converge fast enough. In particular, we need two product rate conditions:

$$\sqrt{n} \cdot \text{RMSE}(\hat{p}) \cdot \text{RMSE}(\hat{m}) \to 0 \qquad \sqrt{n} \cdot \text{RMSE}(\hat{p}) \cdot \text{RMSE}(\hat{\mu}) \to 0$$

which are exactly the product rate conditions we would need if we were just interested in estimating the effect of Z on D and the effect of Z on Y, using the doubly robust estimator.

3.2 Asymptotic Normality and Influence Functions

To perform inference (i.e., compute standard errors and confidence intervals), we need the influence function of the estimator. Since the estimator is a ratio of two means, the influence will look similar to the influence function we used in the ATT setting.

Asymptotic Normality of the DR-LATE Estimator

The asymptotic distribution of the cross-fitted doubly robust LATE estimator is:

$$\sqrt{n}(\hat{\theta}_{\text{LATE}} - \theta) \xrightarrow{d} N(0, \text{Var}(\rho(W)))$$

where $W = (X, Z, D, Y)$ and the influence function ρ for $\hat{\theta}_{\text{LATE}}$ is:

$$\rho(W) = \frac{1}{\mathbb{E}[\psi_D]}(\psi_Y - \theta \cdot \psi_D)$$

with $\theta = \mathbb{E}[\psi_Y]/\mathbb{E}[\psi_D] = \text{LATE}$. The standard error is estimated by

$$\text{s.e.} = \sqrt{\frac{\mathbb{E}_n[\hat{\rho}(W)^2]}{n}}, \qquad \hat{\rho}(W) = \frac{1}{\mathbb{E}_n[\hat{\psi}_D]}(\hat{\psi}_Y - \hat{\theta}_{\text{LATE}} \cdot \hat{\psi}_D)$$

The following pseudocode outlines the full cross-fitting procedure for the DR-LATE estimator, mirroring the implementation in the accompanying Jupyter notebook.

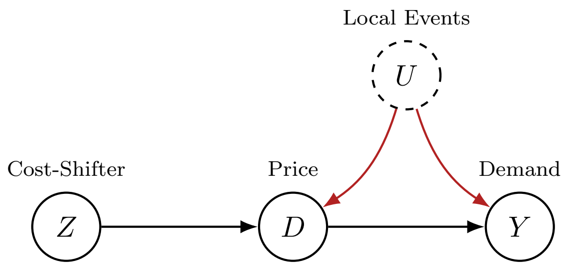
Python: Doubly Robust LATE Estimation

```python
import numpy as np
from sklearn.model_selection import KFold
from sklearn.base import clone

def dr_late(X, Z, D, y, modely0, modely1, modeld0, modeld1,
            modelz, *, nfolds=5, trimming=0.01):
    """
    Doubly Robust LATE estimation with cross-fitting.
    Nuisance models:
      modely0, modely1: E[Y|Z=z, X] for z=0 and z=1 populations
      modeld0, modeld1: E[D|Z=z, X] for z=0 and z=1 populations
      modelz: P(Z=1|X) (propensity score of instrument)
    """
    cv = KFold(n_splits=nfolds, shuffle=True, random_state=123)
    n = X.shape[0]

    # Initialize out-of-fold predictions
    yhat0, yhat1 = np.zeros(n), np.zeros(n)
    Dhat0, Dhat1 = np.zeros(n), np.zeros(n)
    Zhat = np.zeros(n)

    for train, test in cv.split(X):
        # Cross-fit nuisance models for E[Y|X,Z] and E[D|X,Z]
        mask0 = Z[train] == 0
        mask1 = Z[train] == 1
        yhat0[test] = clone(modely0).fit(X[train][mask0], y[train][mask0]).predict(X[test])
        yhat1[test] = clone(modely1).fit(X[train][mask1], y[train][mask1]).predict(X[test])
        Dhat0[test] = clone(modeld0).fit(X[train][mask0], D[train][mask0]).predict_proba(X[test])[:, 1]
        Dhat1[test] = clone(modeld1).fit(X[train][mask1], D[train][mask1]).predict_proba(X[test])[:, 1]

        # Cross-fit propensity score for the instrument
        Zhat[test] = clone(modelz).fit(X[train], Z[train]).predict_proba(X[test])[:, 1]

    # instrument propensity clipping
    Zhat = np.clip(Zhat, trimming, 1 - trimming)

    # Doubly robust scores for numerator and denominator
    yhat = yhat0 * (1 - Z) + yhat1 * Z
    Dhat = Dhat0 * (1 - Z) + Dhat1 * Z
    HZ = Z / Zhat - (1 - Z) / (1 - Zhat)

    psi_Y = yhat1 - yhat0 + (y - yhat) * HZ
    psi_D = Dhat1 - Dhat0 + (D - Dhat) * HZ

    # Point estimate
    theta = np.mean(psi_Y) / np.mean(psi_D)

    # Standard error via influence function
    J = np.mean(psi_D)
    rho = (psi_Y - theta * psi_D) / J
    stderr = np.sqrt(np.mean(rho**2) / n)

    return theta, stderr
```

4 IV with Continuous Treatments: DML-IV

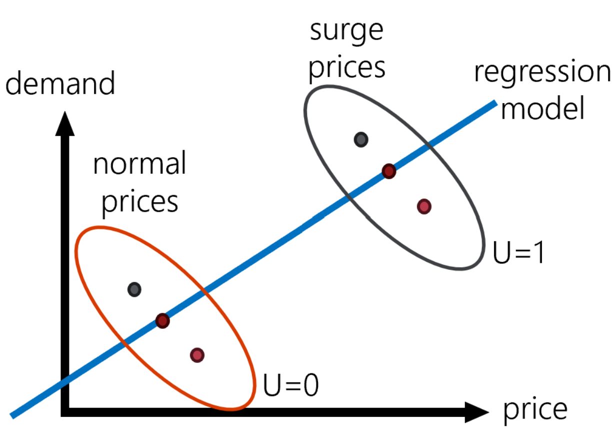
The LATE framework is designed for binary treatments. What if the treatment variable is continuous? For example, in economics, a classic IV application is **demand estimation**, where we want to estimate the causal effect of price (D, continuous) on quantity sold (Y).

This setting is plagued by confounding bias: price is not set exogenously but responds unobserved factors that also influence demand. Unobserved factors that shift the demand curve (e.g., changes in consumer preferences, local events, U) will affect both the price and quantity, creating a confounding path. A common strategy is to find an instrument Z that shifts the *supply* curve (cost of production) but not the demand curve. For example, if we are estimating the demand for coffee, the weather in a coffee-producing country like Colombia could serve as an instrument. Good weather increases supply, lowering the price, but has no direct effect on US consumers' demand for coffee.



4.1 IV Regression: The Visual Intuition

To build intuition for how IV works with continuous treatments, consider a scatter plot of price (D) versus quantity demanded (Y). In the raw data, the observed correlation between price and

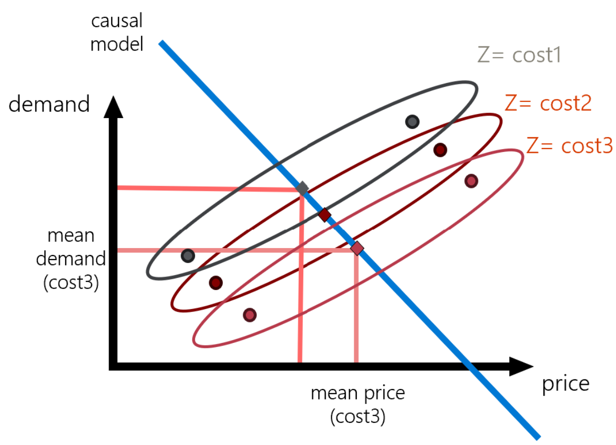


demand is confounded: when local events increase demand, firms respond by raising prices, so we might observe a positive correlation between price and quantity even though the true causal effect of price on demand is negative.

The key insight is that the instrument Z (the cost-shifter) generates variation in price that is *orthogonal* to the demand-side confounders. If we consider a simple linear model $Y = \theta D + U$ where $\mathbb{E}[U] = 0$ and U is uncorrelated with Z, then:

$$\begin{aligned} \mathbb{E}[Y \mid Z] &= \theta \mathbb{E}[D \mid Z] + \mathbb{E}[U \mid Z] \\ &= \theta \mathbb{E}[D \mid Z] + \mathbb{E}[U] \\ &= \theta \mathbb{E}[D \mid Z] \end{aligned}$$

This means that if we plot the conditional mean of Y against the conditional mean of D at different values of the instrument, the slope of that line is the true causal effect θ. The instrument effectively "traces out" the demand curve by shifting the supply curve exogenously.



4.2 The IV Formula via Covariance

With continuous treatments, it is common to assume a partially linear model for the outcome:

$$Y = \theta D + U$$

where θ is the constant, continuous treatment effect we want to estimate. The key IV assumptions are that the instrument Z is conditionally independent of the unobserved confounder U (exogeneity), and that Z does not enter the outcome equation directly (exclusion).

Under these assumptions, we can take the covariance of both sides of the equation with the instrument Z:

$$\text{Cov}(Y, Z) = \text{Cov}(\theta D + U, Z) = \theta \cdot \text{Cov}(D, Z) + \text{Cov}(U, Z)$$

By the exogeneity assumption $\text{Cov}(U, Z) = 0$, this simplifies to:

$$\text{Cov}(Y, Z) = \theta \cdot \text{Cov}(D, Z)$$

Solving for θ gives the famous IV formula for continuous treatments:

The IV (Wald) Estimator for Continuous Treatments

$$\theta = \frac{\text{Cov}(Y, Z)}{\text{Cov}(D, Z)}$$

This formula requires two conditions: (i) **Relevance:** $\text{Cov}(D, Z) \neq 0$, and (ii) **Exclusion:** $\text{Cov}(U, Z) = 0$.

This is the continuous analog of the LATE ratio. It tells us that the treatment effect is the ratio of the covariance of the outcome with the instrument to the covariance of the treatment with the instrument.

4.3 The Analogy to OLS Normal Equations

Another way to think of the IV method is through *moment equations*, which are generalizations of the *normal equations* we saw for OLS. It is instructive to compare the IV moment condition to the one used by Ordinary Least Squares (OLS). OLS solves the normal equation

$$\mathbb{E}[(Y - \theta D) \cdot D] = 0.$$

This equation assumes that the residual $U = Y - \theta D$ is uncorrelated with the treatment D. This fails under confounding, because the unobserved factor U is correlated with D. So we should not be looking for a coefficient θ that makes the residual un-correlated with the treatment D.

The IV approach replaces the problematic moment condition with a new one:

$$\mathbb{E}[(Y - \theta D) \cdot Z] = 0.$$

It assumes the residual $U = Y - \theta D$ is uncorrelated with the *instrument* Z, which is correct. This is a much weaker and more plausible assumption, because the instrument is exogenous by construction. Thus we can think of the IV approach as trying to a coefficient θ, that will make the residual $Y - \theta D$, un-correlated with the instrument Z. Solving for this "normal equation", gives the formula from the previous section (assuming variables have been demeaned):

$$\mathbb{E}[(Y - \theta D) \cdot Z] = 0 \implies \theta = \frac{\mathbb{E}[ZY]}{\mathbb{E}[ZD]} = \frac{\mathrm{Cov}(Z, Y)}{\mathrm{Cov}(Z, D)}$$

4.4 Double/Debiased ML for IV (DML-IV)

To estimate θ in the presence of high-dimensional controls X, we can use the Double/Debiased Machine Learning (DML) framework. Assume the outcome model is partially linear:

$$Y = \theta D + f(X) + U, \quad \mathbb{E}[U \mid X] = 0$$

And the instrument Z is conditionally exogenous given X: $\mathbb{E}[U \mid X, Z] = \mathbb{E}[U \mid X] = 0$.

The key idea is to use Neyman-orthogonal moments, which involves **residualization**. We first estimate the nuisance functions:

- $h(X) = \mathbb{E}[Y \mid X]$
- $m(X) = \mathbb{E}[D \mid X]$
- $r(X) = \mathbb{E}[Z \mid X]$

Then we compute the residuals:

$$\tilde{Y} = Y - h(X) \qquad \tilde{D} = D - m(X) \qquad \tilde{Z} = Z - r(X)$$

Our partially linear model implies that, after residualizing, we have

$$\tilde{Y} = \theta\tilde{D} + \tilde{U}.$$

We cannot simply regress $\tilde{Y}$ on $\tilde{D}$ because $\tilde{D}$ is still correlated with $\tilde{U}$. However, we can use the residualized instrument $\tilde{Z}$ to form a valid moment condition, since $\tilde{Z}$ is un-correlated with $\tilde{U}$:

$$\mathbb{E}[(\tilde{Y} - \theta\tilde{D}) \cdot \tilde{Z}] = 0$$

Solving this moment equation gives the DML-IV estimator.

The DML-IV Estimator

Using cross-fitted estimates of the residuals, $\check{Y}, \check{D}, \check{Z}$, the DML-IV estimator is:

$$\hat{\theta}_{\text{DML-IV}} = \frac{\mathbb{E}_n[\check{Y}\check{Z}]}{\mathbb{E}_n[\check{D}\check{Z}]}$$

where $\check{Y}, \check{D}, \check{Z}$ are the cross-fitted residuals from regressing each variable on X.
This estimator is $\sqrt{n}$-consistent and asymptotically normal

$$\sqrt{n}(\hat{\theta}_{\text{DML-IV}} - \theta) \to N(0, \text{Var}(\psi(W))$$

assuming that the estimates $\hat{h}, \hat{m}, \hat{r}$ achieve fast enough RMSE rates, i.e.

$$\sqrt{n}\text{RMSE}(\hat{r})^2 \to 0 \qquad \sqrt{n}\text{RMSE}(\hat{h})\text{RMSE}(\hat{r}) \to 0, \qquad \sqrt{n}\text{RMSE}(\hat{m})\text{RMSE}(\hat{r}) \to 0$$

Its influence function is:

$$\psi(W) = \frac{(\tilde{Y} - \theta\tilde{D})\tilde{Z}}{\mathbb{E}[\tilde{D}\tilde{Z}]}$$

and the standard error can be estimated using the empirical analogue of this variance:

$$\text{s.e} = \sqrt{\frac{\mathbb{E}[\hat{\psi}(W)^2]}{n}} \qquad \hat{\psi}(W) = \frac{(\check{Y} - \theta\check{D})\check{Z}}{\mathbb{E}_n[\check{D}\,\check{Z}]}$$

Limits of DML-IV Method

The DML-IV methodology applies verbatim under the slightly weaker assumptions that:

$$\mathbb{E}[Y \mid D, X, U] = \theta D + g(X, U)$$

and $U \perp\!\!\!\perp Z \mid X$. This is a more permissive structural equation framework, as it allows for a more general form of structural equations:

$$Y(d, x, u) = f(\epsilon_Y)\, d + h(x, u, \epsilon_Y)$$

The pseudocode below implements the DML-IV estimator with cross-fitting, following the implementation in the accompanying Jupyter notebook.

Python: DML for Partially Linear IV

```python
import numpy as np
from sklearn.model_selection import KFold, cross_val_predict

def dml_iv(X, Z, D, y, modely, modeld, modelz, *, nfolds=5):
    """
    DML for the Partially Linear IV model with cross-fitting.
    Nuisance models:
      modely: for E[Y|X]
      modeld: for E[D|X]
      modelz: for E[Z|X]
    """
    cv = KFold(n_splits=nfolds, shuffle=True, random_state=123)
    n = X.shape[0]

    # Get out-of-fold predictions for nuisance functions
    y_hat = cross_val_predict(modely, X, y, cv=cv)
    d_hat = cross_val_predict(modeld, X, D, cv=cv)
    z_hat = cross_val_predict(modelz, X, Z, cv=cv)

    # Compute residuals
    res_y = y - y_hat
    res_d = D - d_hat
    res_z = Z - z_hat

    # Point estimate from residual-on-residual IV regression
    theta = np.mean(res_y * res_z) / np.mean(res_d * res_z)

    # Standard error via influence function
    epsilon = res_y - theta * res_d
    J = np.mean(res_d * res_z)
    psi = epsilon * res_z / J
    stderr = np.sqrt(np.mean(psi**2) / n)

    return theta, stderr
```

5 The Problem of Weak Instruments

A critical assumption for IV estimation is **relevance**: the instrument Z must have a causal effect on the treatment D, i.e., $\mathrm{Cov}(\tilde{D}, \tilde{Z}) \neq 0$ in the partially linear setting, or $\mathrm{ATE}(Z \to D) \neq 0$ in the LATE setting. If this quantity is very close to zero, we have a **weak instrument**. This is a major practical problem in applied work.

When the instrument is weak, the denominator of the IV estimator is close to zero. This has two severe consequences:

1. **High Variance:** The estimator becomes very noisy and sensitive to small changes in the data.

2. **Non-Normal Distribution:** The sampling distribution of the IV estimator is no longer well-approximated by a normal distribution, even in moderately large samples. This means that standard t-tests and confidence intervals are invalid.

5.1 Detecting Weak Instruments: The F-Test

How can we know if our instrument is weak? The standard diagnostic is to look at the first-stage regression of the treatment on the instrument and controls. After residualizing out the controls X, the first-stage equation is:

$$\tilde{D} = \pi_0 \tilde{Z} + \epsilon$$

where $\tilde{D} = D - \mathbb{E}[D \mid X]$ and $\tilde{Z} = Z - \mathbb{E}[Z \mid X]$ are the residualized treatment and instrument, respectively. The null hypothesis of a weak instrument test assumes π_0 takes a very small value, which induces a bias in the second-stage IV estimate $\hat{\theta}$.

The First-Stage F-Test

A frequently used heuristic for the bias induced by a weak instrument is the **Nagar bias**, which posits that for a true first-stage parameter π_0, an estimate $\hat{\pi}$ with variance σ_π^2 induces a bias on the second-stage estimate of approximately $\tau = 1/\lambda^2$, where $\lambda^2 = \pi_0^2/\sigma_\pi^2$. The null hypothesis of a weak instrument is that this bias is large, i.e., $\lambda^2 \leqslant 1/\tau^*$ for some user-specified tolerance τ^* (e.g., $\tau^* = 0.1$).
To check this, we use a heteroskedasticity-robust F-statistic:

$$F = \frac{\hat{\pi}^2}{\hat{\sigma}_\pi^2}$$

where $\hat{\pi} = \mathbb{E}_n[\check{D}\check{Z}]/\mathbb{E}_n[\check{Z}^2]$ is the OLS estimate from the residualized first-stage regression (with estimated treatmend and instrument residuals) and $\hat{\sigma}_\pi^2$ is its heteroskedasticity-robust estimated variance of the OLS, i.e.

$$\hat{\sigma}_\pi^2 = \frac{1}{n}\frac{\mathbb{E}_n[(\check{D} - \hat{\pi}\check{Z})^2\check{Z}^2]}{\mathbb{E}_n[\check{Z}^2]^2}.$$

Under the assumption of asymptotic normality of $\hat{\pi}$, we can deduce that

$$\frac{\hat{\pi} - \pi_0}{\hat{\sigma}_\pi} \xrightarrow{d} N(0, 1)$$

and we have:

$$\frac{\hat{\pi}}{\hat{\sigma}_\pi} \approx N\left(\frac{\pi_0}{\sigma_\pi}, 1\right)$$

Under the null hypothesis of a weak instrument, the F-statistic

$$\text{F-statistic} \triangleq \hat{\pi}^2/\hat{\sigma}_\pi^2$$

follows approximately a non-central $\chi^2_{1,c}$ distribution with centrality parameter $c = \lambda^2 \leqslant 1/\tau^*$ (this is simply the distribution of the square of a Gaussian with mean c and variance 1). With confidence α, we can reject the null (conclude the instrument is strong) if F is larger than the $1 - \alpha$ percentile of the $\chi^2_{1,1/\tau^*}$ distribution.

Rule of Thumb: A common heuristic, proposed by Staiger and Stock (1997), is that if the first-stage **F-statistic is greater than 10**, we can be confident that the instrument is not weak.

5.2 Robust Inference: The Anderson-Rubin Statistic

If the F-statistic is low, we cannot trust our standard confidence intervals. What can we do? We need an inference procedure that is robust to weak instruments. The key idea, first proposed by Anderson and Rubin (1949), is to rearrange the moment condition.

Recall the IV moment condition: $\mathbb{E}[(\tilde{Y} - \theta\tilde{D}) \cdot \tilde{Z}] = 0$. Let us test a specific null hypothesis, $H_0 : \theta = \theta_0$. Under this null, the transformed outcome variable $\tilde{Y} - \theta_0\tilde{D}$ should be uncorrelated with the instrument Z, because $\tilde{Y} - \theta_0\tilde{D} = \tilde{U}$, and $\tilde{Z}$ is uncorrelated with both $\tilde{U}$. This gives us a valid condition that does not involve dividing by the first-stage covariance:

$$\mathbb{E}[(\tilde{Y} - \theta_0\tilde{D}) \cdot \tilde{Z}] = 0$$

This equation holds even if the instrument is completely irrelevant. We can form a test statistic based on this moment. By testing a range of possible values for θ_0 and collecting all the values that we fail to reject, we can construct a valid confidence set for the true θ.

5.3 The $C(\theta)$ Statistic

For any hypothesized value θ_0, we can form the moment using our cross-fitted residuals. In the DML-IV setting, the moment for a given θ_0 is:

$$\check{M}(\theta_0) = \mathbb{E}_n[(\tilde{Y} - \theta_0\tilde{D})\tilde{Z}]$$

Under the null hypothesis $H_0 : \theta = \theta_0$, the expectation of this moment is zero. The $C(\theta_0)$ statistic is a test for whether the sample mean of this moment is significantly different from zero:

$$C(\theta_0) = n \cdot \frac{\check{M}(\theta_0)^2}{\check{\Omega}(\theta_0)}$$

where $\check{\Omega}(\theta_0) = \mathrm{V}_n[(\tilde{Y} - \theta_0\tilde{D})\tilde{Z}]$ is the empirical variance of the moment. Under the null hypothesis, and since $\check{M}(\theta_0)$ is a mean-zero empirical average, we have:

$$\sqrt{n}\frac{\check{M}(\theta_0)}{\sqrt{\check{\Omega}(\theta_0)}} \to_d N(0,1) \tag{1}$$

Thus the statistic $C(\theta_0)$, which is the square of this quantity follows a χ^2_1 distribution with 1 degree of freedom, *regardless of the strength of the instrument* (this is simply the distribution of the square of a standard mean-zero Gaussian random variable). This is the crucial property that makes this approach robust to weak instruments.

Many instruments and treatments

When we have many instruments and treatments, we are solving a vector of equations. Thus $\check{M}(\theta_0)$ is a $m = \dim(Z)$ vector of conditions and $\check{\Omega}(\theta_0)$ will be an empirical covariance matrix of this vector. We have that under the null hypothesis, the quantity:

$$\sqrt{n}\check{\Omega}^{-1/2}(\theta_0)\check{M}(\theta_0) \to_d N(0, I_m).$$

Thus the squared ℓ_2-norm of this scaled vector, i.e.

$$\|\sqrt{n}\check{\Omega}^{-1/2}(\theta_0)\check{M}(\theta_0)\|_2^2 = n\check{M}(\theta_0)\check{\Omega}^{-1}(\theta_0)\check{M}(\theta_0)$$

will be approximately distibuted as the sum of the squares of the coordinates of an m-dimensional standard Gaussian random variable. This distribution is called the χ^2_m distribution, the chi-squared distribution with m degrees of freedom. Thus we can use this statistic, i.e., $n\,\check{M}(\theta_0)\check{\Omega}^{-1}(\theta_0)\check{M}(\theta_0)$ and check it against the $1-\alpha$ quantile of the χ^2_m distribution, in order to reject a candidate solution θ_0.

5.4 Confidence Set by Inverting the Test

We can now build a valid confidence set by inverting this test. We test a grid of possible values for θ_0. For each value, we compute $C(\theta_0)$ and compare it to the critical value from a χ^2_1 distribution (e.g., 3.84 for a 95% confidence level).

Weak-IV Robust Confidence Set

The $(1-\alpha)$ confidence set for θ is the set of all values θ_0 for which the test fails to reject:

$$\mathrm{CR}_{1-\alpha} = \{\theta_0 \in \mathbb{R} \;:\; C(\theta_0) \leqslant c_{1-\alpha}\}$$

where $c_{1-\alpha}$ is the $(1-\alpha)$ percentile of the $\chi^2(m)$ distribution.

This procedure gives a valid confidence set for θ even if the instrument is very weak. The resulting set might not be a simple, single interval; it could be the union of disjoint intervals, or it could even be the entire real line if the data is completely uninformative. But it correctly reflects the true uncertainty in our estimate.

5.5 Pseudocode for Weak-IV Robust Inference

Python: Weak-IV Robust Confidence Set

```python
import numpy as np
import scipy.stats

def weak_iv_confidence_set(check_y, check_d, check_z,
                           grid, alpha=0.05):
    """
    Constructs a weak-IV-robust confidence set by
    inverting the C(theta) test.

    Parameters:
      check_y, check_d, check_z: cross-fitted residuals
      grid: array of theta_0 values to test
      alpha: significance level (e.g. 0.05 for 95% CI)
    """
    n = check_y.shape[0]
    crit_val = scipy.stats.chi2.ppf(1 - alpha, df=1)

    accept_region = []
    for theta_0 in grid:
        moment = (check_y - theta_0 * check_d) * check_z
        c_theta = n * np.mean(moment)**2 / np.var(moment)
        if c_theta <= crit_val:
            accept_region.append(theta_0)

    return accept_region
```

Is the point estimate still the ratio in this weak instrument regime?

Yes, the point estimate itself is still the ratio $\hat{\theta} = \mathbb{E}_n[\tilde{Y}\tilde{Z}]/\mathbb{E}_n[\tilde{D}\tilde{Z}]$ (for DML-IV) or $\hat{\theta} = \mathbb{E}_n[\psi_Y]/\mathbb{E}_n[\psi_D]$ (for DR-LATE). The procedure described here is for building a more reliable *confidence interval* around that estimate, which is especially important in the weak instrument case. The point estimate can be very noisy in such regimes, and you should expect the corresponding confidence interval to be very large, correctly reflecting that uncertainty.

6 In-Class Activity: Spot the Instrument

To solidify the concepts from this chapter, we work through two applied vignettes that require identifying the instrument, treatment, and outcome, and reasoning about the IV assumptions.

6.1 Vignette D: Health Insurance and Emergency Room Visits

A state government wants to know whether expanding Medicaid coverage reduces emergency room (ER) visits. People who sign up for Medicaid may be systematically sicker or poorer than those who do not, creating a selection bias problem. In Oregon in 2008, the state expanded Medicaid through a lottery: eligible residents could enter a lottery, and winners were given the opportunity to apply for Medicaid.

- **Instrument (Z):** Winning the Medicaid lottery.
- **Treatment (D):** Enrolling in Medicaid.
- **Outcome (Y):** Number of ER visits.
- **Exclusion Restriction:** Winning the lottery affects ER visits *only* through its effect on Medicaid enrollment. This is plausible because the lottery itself is just a piece of paper; it has no direct health effects.
- **Compliers:** People who enrolled in Medicaid because they won the lottery, but who would not have enrolled otherwise. Not all lottery winners enrolled (some did not complete the application), and some lottery losers obtained insurance through other means.

This is the famous Oregon Health Insurance Experiment. A notable finding from Finkelstein et al. (2012) was that Medicaid actually *increased* ER visits, contrary to the policy expectation. The LATE applies to the complier subpopulation—those on the margin of enrollment.

6.2 Vignette E: Online Recommendations and User Engagement

A travel website wants to estimate the causal effect of users becoming premium members (D) on the number of days they visit the site (Y). Users who choose to become premium members are likely already more engaged, creating a selection bias. The company runs an A/B test: a random subset of users sees an "easier signup" flow (fewer clicks, prominent button), while the control group sees the standard flow.

- **Instrument (Z):** The signup flow assignment (easy vs. standard).
- **Treatment (D):** Whether the user becomes a premium member.
- **Outcome (Y):** Number of days visiting the site.
- **Exclusion Restriction:** The signup flow design affects site visits *only* through its effect on premium membership take-up. This requires that the flow itself does not directly change browsing behavior.
- **Compliers:** Users who signed up for premium because the flow was easier, but who would not have signed up under the standard flow.

7 Summary and Key Takeaways

This chapter introduced instrumental variables, a cornerstone of modern causal inference for handling unobserved confounding. The key idea is to find a source of exogenous variation—the instrument—that affects the treatment but does not directly affect the outcome.

1. **IV for Unobserved Confounding.** Instrumental Variables are a powerful strategy for identification when conditional exogeneity fails due to unobserved confounders.

2. **LATE for Compliers.** For binary treatments, IV identifies the Local Average Treatment Effect (LATE)—the ATE for the subpopulation of *compliers* whose treatment status is changed by the instrument. This requires the monotonicity (no defiers) assumption.

3. **Assumptions Matter.** The LATE interpretation relies on the exclusion restriction and monotonicity. The IV strategy for continuous treatments relies on a partial linearity assumption.

4. **DR and DML for IV.** We can use our familiar Doubly Robust (for LATE) and Double ML (for continuous IV) machinery to get robust and efficient estimates with high-dimensional controls.

5. **Weak Instruments Require Robust Inference.** When the first stage is weak, standard IV confidence intervals are unreliable. The first-stage F-statistic is the primary diagnostic ($F > 10$). When instruments are weak, we must use methods like inverting the $C(\theta)$ statistic to get valid confidence sets.

Instrumental variables represent a powerful tool, but their validity hinges on strong, untestable assumptions (especially the exclusion restriction). The search for a valid instrument is one of the most creative and challenging aspects of applied causal inference.

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